

Rahnuma

रहनुमा رہنما

A working guide to the ILM, PANINI and Zistgah stack, from bytes and scripts to compilers, Pāṇini and quantum substrates

प्रकृति

	written	Devanagari hub	Romenagri	and back
Devanagari	प्रकृति	→ प्रकृति	→ prak_ri_ti	→ प्रकृति
Telugu	ప్రకృతి	→ प्रकृति	→ prak_ri_ti	→ ప్రకృతి
Bengali	প্রকৃতি	→ प्रकृति	→ prak_ri_ti	→ the same identity
Urdu, as usually written	ہندوی	→ हनदवी	→ _hana_dawee	→ हनदवी
Urdu, with zer and sukun	ہندویٰ	→ हिन्दवी	→ _hin_dawee	→ हिन्दवी

One identity, many scripts. Telugu and Bengali meet Devanagari in one hub, become the same Romenagri word, and come back exactly. Urdu meets the hub at what is written: without its short-vowel marks the word reads hanadawee; with zer and sukun it is hindawi. Every cell here was produced by the Hindawi tools while this guide was built.

Abhishek Choudhary, AyeAI. Version 2.0.0, 3 October 2026.

Rahnuma, the edition for: Academic researcher

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Part I: Orientation

3. Start here: find your way in

In plain words. Four steps to find your part of the guide: who you are, your board or field, what you want to do, and your path with a PDF made for you.

This guide serves a school student and a senior researcher; a village health worker and an investor. Nobody needs all of it. Four steps find your part, on the site or with this page:

1. **Who you are.** A school student, a college student, a postgraduate or doctoral researcher; a teacher; an academic researcher; a working professional, someone in public service, policy or law, someone in community, NGO or village work, an artist, writer or maker; an industry partner or investor; or an AI agent working for a person.
2. **Your board or your field.** A school student chooses a board, ICSE, ISC, CBSE, IGCSE, IB, a state board or another; and a class. Everyone else chooses a field from the international standard classification of fields of study, ISCED-F 2013, the same one FAKIR's lattice uses. It places each subject where its own scholars would: political science is under social and behavioural sciences, not the humanities; law is under business, administration and law; linguistics is under languages.
3. **What you want to do.** Learn the basics, run the software, study or research, teach, contribute, or adopt, fund or invest.
4. **Your path and your edition.** The site lists your chapters in order, with reading times and the components to try first, and offers a PDF edition made for you. Every edition below is a printed selection from this one guide, so all of them stay in step.

Edition	PDF
School student	role-school.pdf
College student	role-college.pdf
Postgraduate or doctoral researcher	role-pg.pdf
Teacher or lecturer	role-teacher.pdf
Academic researcher	role-researcher.pdf
Working professional or engineer	role-professional.pdf
Public service, policy or law	role-public.pdf
Community, NGO or village work	role-community.pdf
Artist, writer or maker	role-creator.pdf
Industry partner, investor or funder	role-industry.pdf
Generic programmes and qualifications	field-00.pdf
Education	field-01.pdf
Arts and humanities	field-02.pdf
Social sciences, journalism and information	field-03.pdf

Edition	PDF
Business, administration and law	field-04.pdf
Natural sciences, mathematics and statistics	field-05.pdf
Information and Communication Technologies (ICTs)	field-06.pdf
Engineering, manufacturing and construction	field-07.pdf
Agriculture, forestry, fisheries and veterinary	field-08.pdf
Health and welfare	field-09.pdf
Services	field-10.pdf

The first public version of this guide, written for advanced researchers and senior professionals, stays as it was at [version 1](#).

Part II: One architecture

4. Where the work comes from

In plain words. This work began long before today's AI. It grew along five connected lines: languages and computers, learning machines, robots and medicine, organised knowledge, and the future of people and their homes.

This work did not begin with today's language models, and it is not a collection of separate projects. It is one architecture built along several lines of work that started at different times, and it reads correctly only in that order. The dated record, with the place where each entry can be checked, is the next chapter; this one gives the lines themselves.

Five lines that intersect

Linguistic and computational identity. Romenagri (2003), the reversible ASCII-7 kernel, came first; the Hindawi Programming System (15 August 2004) built a complete systems-programming stack on it; ILM, Integrative Linguistic Multiscript, succeeds Hindawi and generalises it to every script and language; PANINI's construct model now sits above ILM as the common core of programming, prompting and realization.

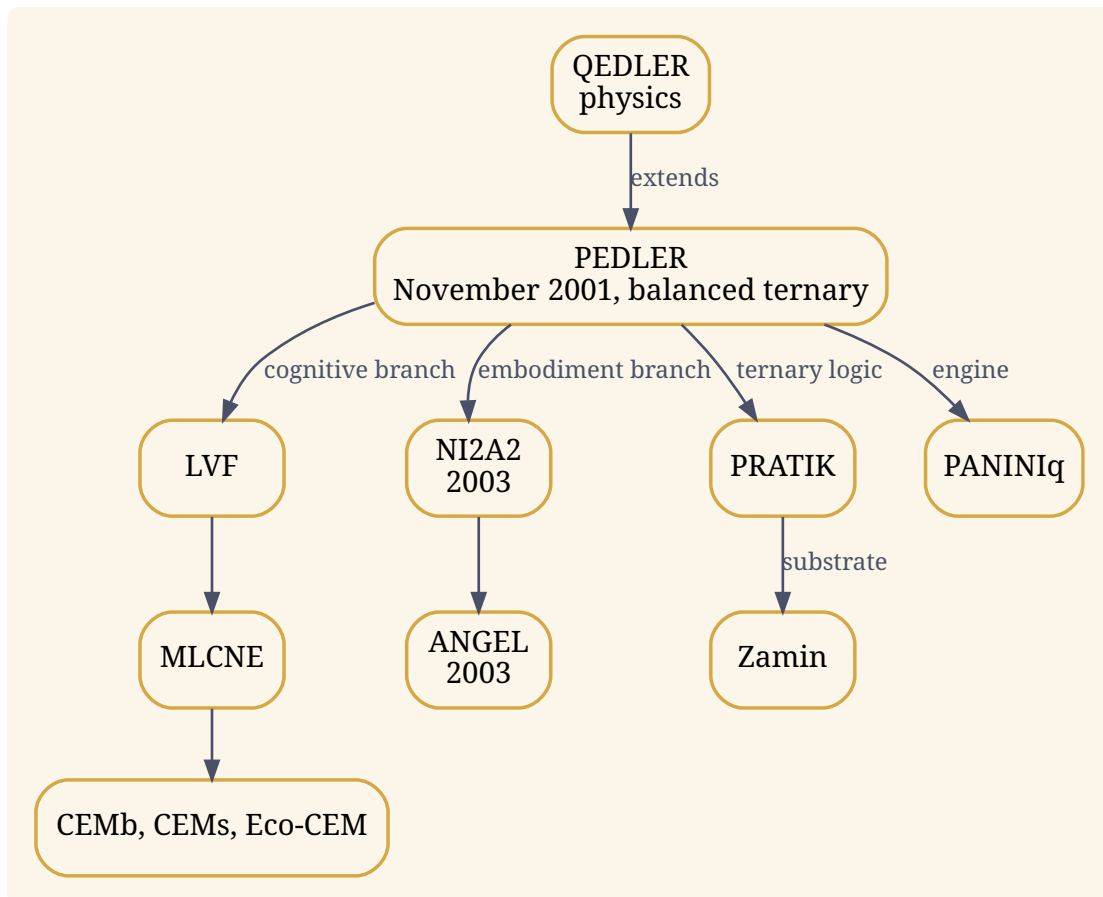


Figure 1. PEDLER's branches: cognition, embodiment, physics and silicon.

Cognition and event-driven computation. PEDLER, the Point Event-Driven Learner (November 2001, Indian patent application 3033/CHE/2011), is not one genealogy but two branches: a cognitive branch through LVF and MLCNE to the cognitive enablement modules, CEMb and CEMs, and an embodiment

branch through NI2A2 (2003) to the cognitive robot ANGEL (2003). QEDLER carries PEDLER into physics. PRATIK and Zamin take its balanced-ternary, event-driven logic towards silicon.

Embodiment, medicine and field systems. HMSEI, a medical wearable (2002); ANGEL; TARA; RDK, among Nokia's global top ten in 2010; Dr Rho, the TARA and RDK medical telepresence platform, Highly Commended at the IET Innovation Awards in 2014; GramSheel's Village Knowledge Center; a Stockholm Challenge finalist in 2010. The AyeAI Triad closes this line formally: AyeAM for embodiment, AyeAI for cognition, AyeCNSe for coordination and communication.

Knowledge and reasoning. UKOP, the reference architecture for human knowledge across ISIC, ISCO and ISCED; FAKIR, its kernel; Dhancha, the spine every domain is built on; the domains themselves; and the working layer that runs on them: GENIE, the cyclers, AAB and Mez.

Continuity and civilisation. Humanesque, its constitutional derivation from the Proclamation of Individual Equity, TransEg, the transferred alter ego, and the habitats on Earth, the Moon and Mars.

The lines cross constantly. PEDLER supplies the Act, the executable intent that the constitutional corpus is built from; Romenagri and ILM carry every language into FAKIR's lattice; PANINI runs the cyclers and also drives the realization compilers; TransEg receives what yadein records. That is why the chapters that follow teach the pieces as nodes of one graph rather than as a list.

The invariant

† What the architecture preserves, whatever changes around it, is meaning, intent, identity, agency, capability, provenance, equity and continuity, while the representation, the language, the script, the formalism, the machine, the substrate, the embodiment, the social setting and eventually the habitat all change. Romenagri is the smallest instance of the principle: a name keeps its identity through every tool of a toolchain. TransEg is the largest: a person's constituted identity keeps its continuity beyond the substrate that first carried it.

Sovereignty and equity

The Proclamation of Individual Equity, PoIE ([zistgah/poie](https://doi.org/10.5281/zenodo.21397274), doi:10.5281/zenodo.21397274), makes the individual the primary locus of equity, identity and agency. Sovereignty is recursive from there: individual, family, community, institution, province, nation and beyond, across political, economic, social and technological dimensions. Technological sovereignty, in this sense, is the capacity to determine, govern, develop, reproduce, modify and deploy one's own domain; the precise, testable form of it for software is given in Chapter 21. The aim that follows is enablement: that people become producers, builders, researchers and owners of computing in their own languages, not permanent consumers of systems built elsewhere.

The orientation map

His poster "The Expanse of Our Epistemic Realms" runs one axis from the inner to the outer: being, meaning, knowledge, language, cognition, code, systems, society, civilisation, planet. At its apex stands Kaivalyik AGI; beneath it, the Equitable Mind, one humanity with one identity and infinite

expressions; and the rule that every sentient individual, bioform or synthetiform, is a sovereign. Read the rest of this Part along that axis: language and identity first, then computation, cognition, knowledge and agency, realization, and the Acts that carry them to civilisational scale.

6. Architecture: core, projections, layers

In plain words. One shared core, Humanesque, is seen through three cultures: Indian, Persian and Western. Around it sit the parts that turn an idea, in any language, into something real.

One core, three projections

Humanesque is the core: one shared, multidimensional, executable typed hypergraph, with a federated ontology of 26 positions, O_0 to O_{25} , a common execution and evidence spine, and components that can be implemented independently. Its GitHub organisation is hmnsq, where every core stem repository is to live.

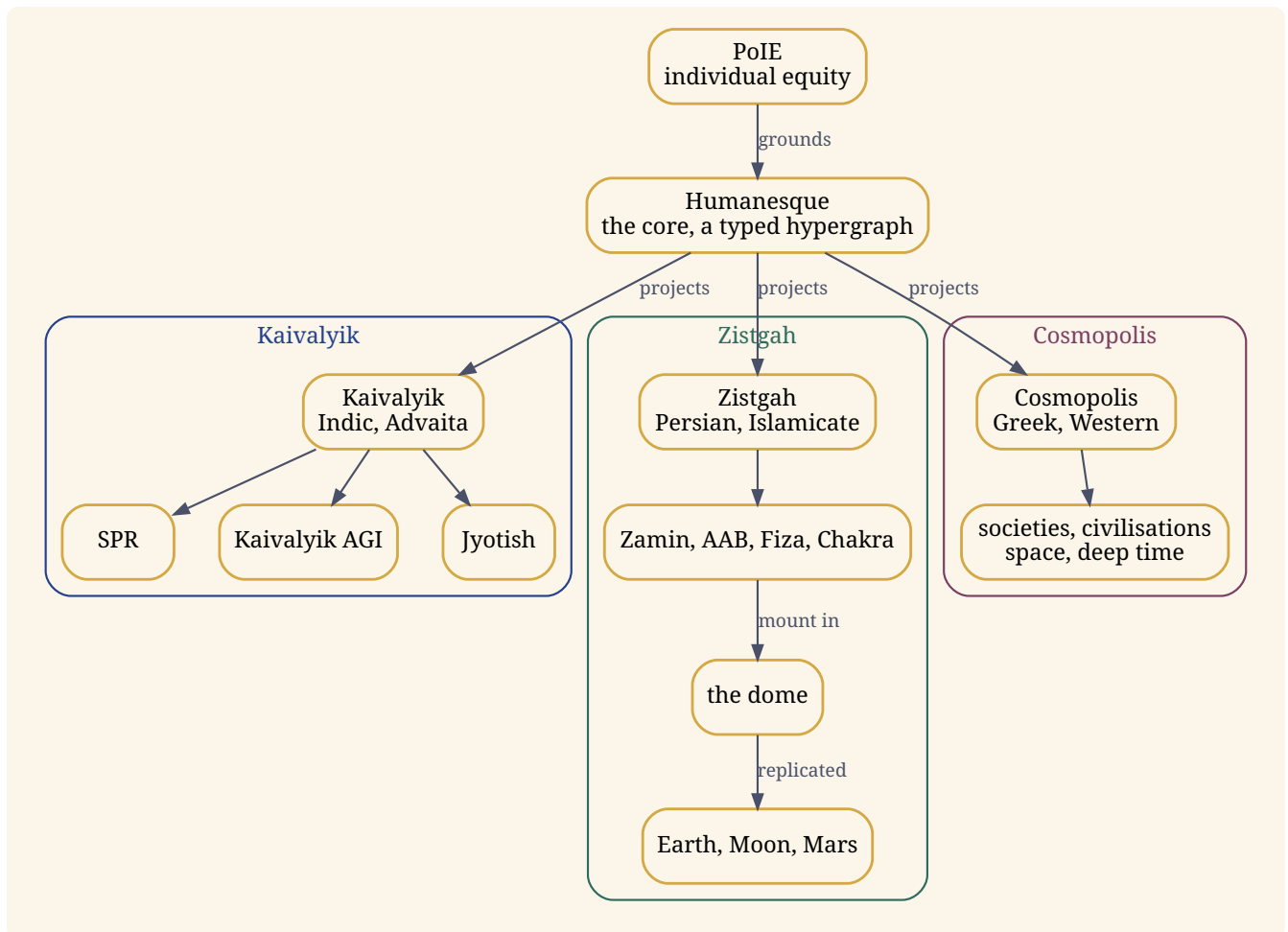


Figure 2. Humanesque and its projections: parallel cultural traversals, not stages.

Kaivalyik, Zistgah and Cosmopolis are three projections over it, running in parallel: the Indic, including the Advaita framing; the Persian and Islamicate; and the Greek, Western and classical. A projection is a domain-specific view and traversal of the whole architecture, with its own culturally sensitive naming and branding. The projections are not stages, not levels of a hierarchy and not versions of one another; they intersect, diverge and reconverge, and none completes another. Zistgah is not the parent of the other two. Their organisations are kaivalyik, zistgah and c-polis; today zistgah carries the working repositories.

The PANINI stack

Layer	What it holds	Where to look
Front end	The languages a program or a prompt can arrive in, Sanskrit first: ILM and the language parsers	project-ilm/ilm.codes, zistgah/tajziya
Middleware	PANINI's own front ends, from <code>ada.pni</code> to <code>zig.pni</code> and <code>sysml.pni</code> , and the PANINI language itself	zistgah/panini, zistgah/humanesque
Core PANINI	The construct model: common semantics plus decorators	zistgah/humanesque, <code>ilm/constructs.csv</code> and <code>ilm/decorators.csv</code>
Realization backends	Where a program meets a substrate: host toolchains for code; PANINIq for oscillator and quantum substrates; PANINIB for biology; PANINIphy for physical systems	zistgah/paniniq, zistgah/paninib, zistgah/paniniphy

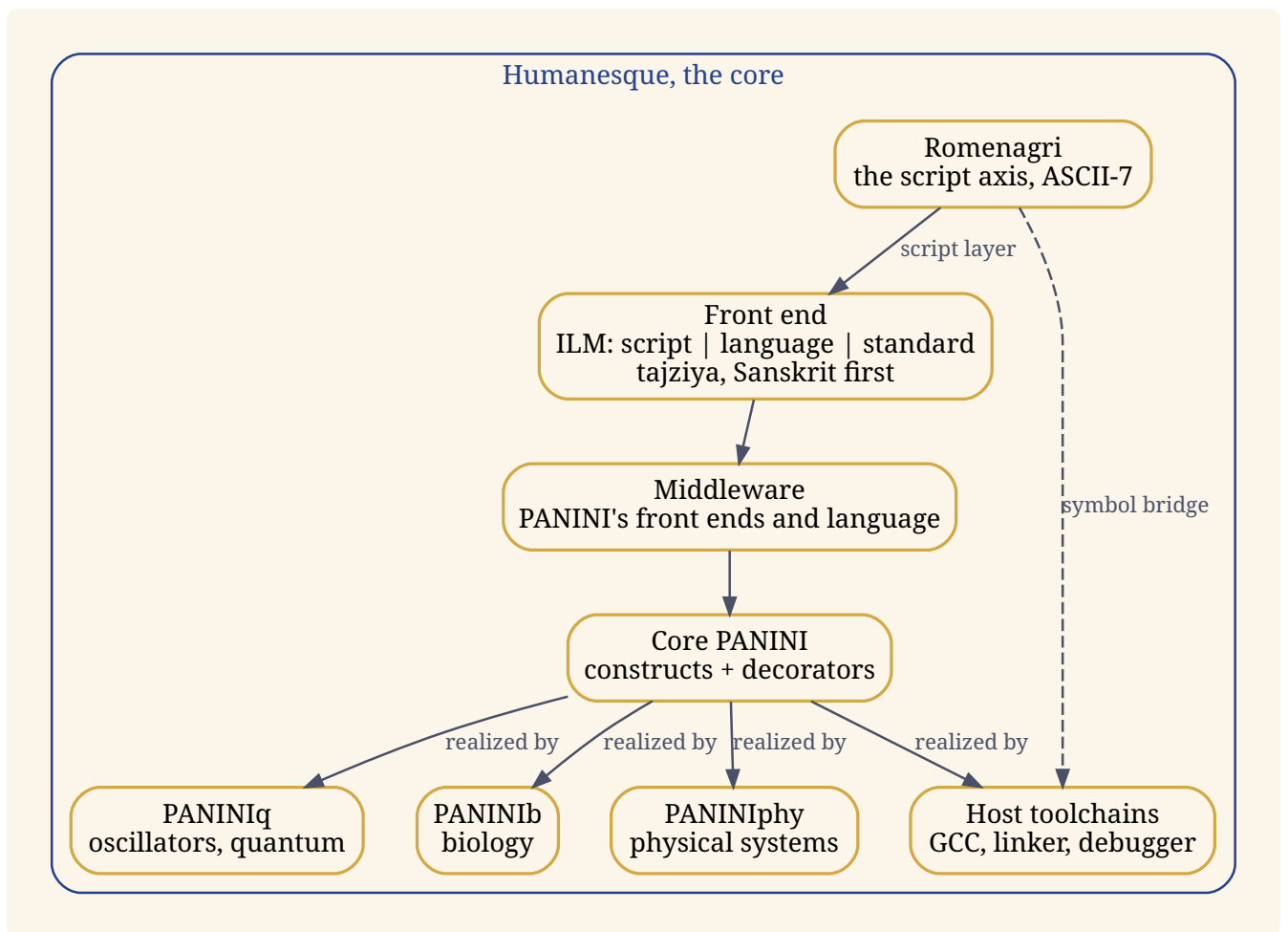


Figure 3. The PANINI stack, with Romenagri as the script layer beneath every front end.

Beneath every front end lies the script layer, Romenagri. Every keyword and every name of a program becomes a word over A to Z, a to z and the underscore, reversibly, so that the whole existing toolchain, from lexer to linker to debugger, carries it unchanged, and the inverse renders it back in the script wherever a person reads it. Chapters 18, 19 and 21 explain it and run it.

A construct such as a counted loop is defined once in the core and translated once per human language. A decorator carries what a particular host language drags in with it, such as `stdio.h` for printing in C. Core PANINI, in his words, “does not violate anything”; the front ends and the realization backends are all projections of it.

How the parts compose

Humanesque	the core; Kaivalyik, Zistgah and Cosmopolis project it in parallel; everything below lives in it, nothing below leads to it
front end	ILM (script language standard) and tajziya, Sanskrit first the script axis is Romenagri, which persists as the symbol bridge through compiler, linker, ELF, DWARF and debugger
middleware	PANINI's own front ends and the PANINI language
core	core PANINI: construct model = common semantics + decorators
realization	host toolchains for code; PANINIq (oscillator, quantum); PANINIb (biology, down to sequence IR); PANINIphy (physical systems); PRATIK and Zamin as a balanced-ternary substrate

Around the stack: the cyclers are written in PANINI and run by one engine; GENIE composes and runs research cycles; AAB paints tasks and runs the process cyclers; FAKIR, UKOP and Dhancha supply the domain lattice and its invariants; Mez, the Cognitive Workbench, composes any of them for an exercise without absorbing them; the provenance systems seal every artifact; VGC gates the work between people and AI. Two readings must be kept apart. As genealogy, the order is Romenagri, Hindawi, ILM, PANINI and the realization arms. As architecture, Romenagri sits inside ILM's script axis, the cyclers are written in PANINI rather than downstream of it, and Humanesque is the core that contains the whole, not its end point.

The realization spine

The realization repositories state the spine they share. PANINI lowers declarative intent through language, semantic IR, a realization requirement, domain resolution and a domain IR, and then a common layer of geometry, trajectory, animation, observation, verification and an ArtifactGraph. PANINIb lowers biological intent through semantic, resolution, CISC, RISC and sequence IRs and emits two coupled artifacts, a physical realization plan and a digital execution run; PANINIphy lowers physical intent through a physical IR and physical-domain engines. “IR contracts are the product. Laboratories and machines are external adapters.” And the same repositories keep the states of a result apart: desired, predicted, observed and validated phenotype are four different things, as are compilation success, execution success, target satisfaction and biological validation.

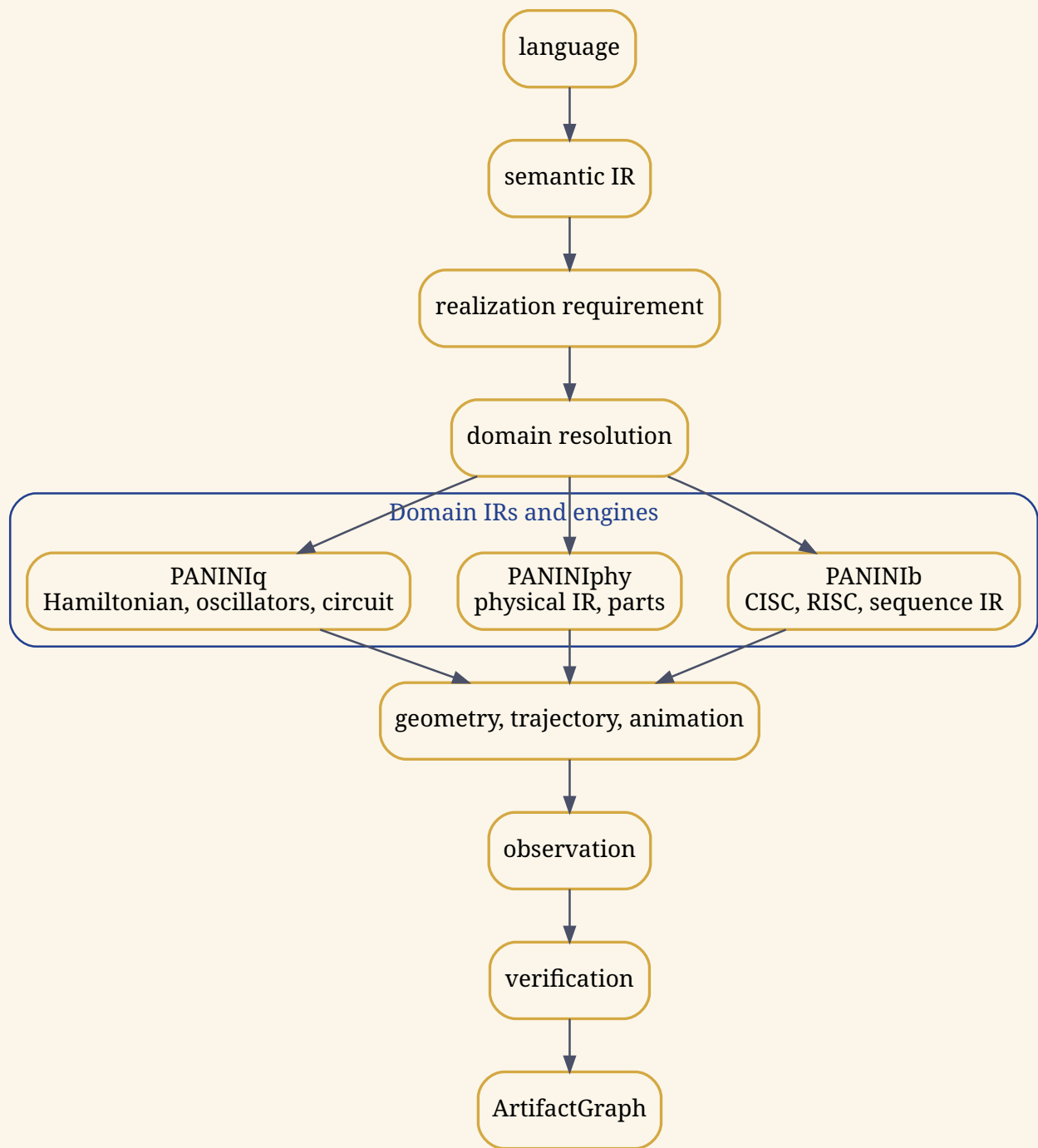


Figure 4. The realization spine the realization arms share.

The habitat elements

Zamin is the ground: the balanced-ternary hardware of PRATIK. AAB is the water: the paint program for systems, from painted intent to verified, sealed artifact, where the verification-gated quests run. Fiza is the air: the environmental replica. Chakra is the turning sky: the observatory and its calendars. All of them mount in one shared virtual dome, which the habitats on the Moon and Mars replicate.

The cyclers

A cycler is a sequence of prompts and the outputs they produce, written down so that it can be edited, shared and run with any AI. Six are classified by what they produce: matba (print), khwab (visual), awaz (audio), tilasm (immersive), pench (embodied) and yadein (the record); genie runs research cycles. Each has its own purpose, contract, context, state model, invariants, failure modes, evidence requirements, workflow and artifact model; only the engine is shared. The protocol behind every cycle, in the author's words: intent, then context, then a meaningful prompt; any AI answers and the human inspects; the artifacts and responses are kept under configuration management; the next prompt follows, until a final artifact that the human authored with intention.

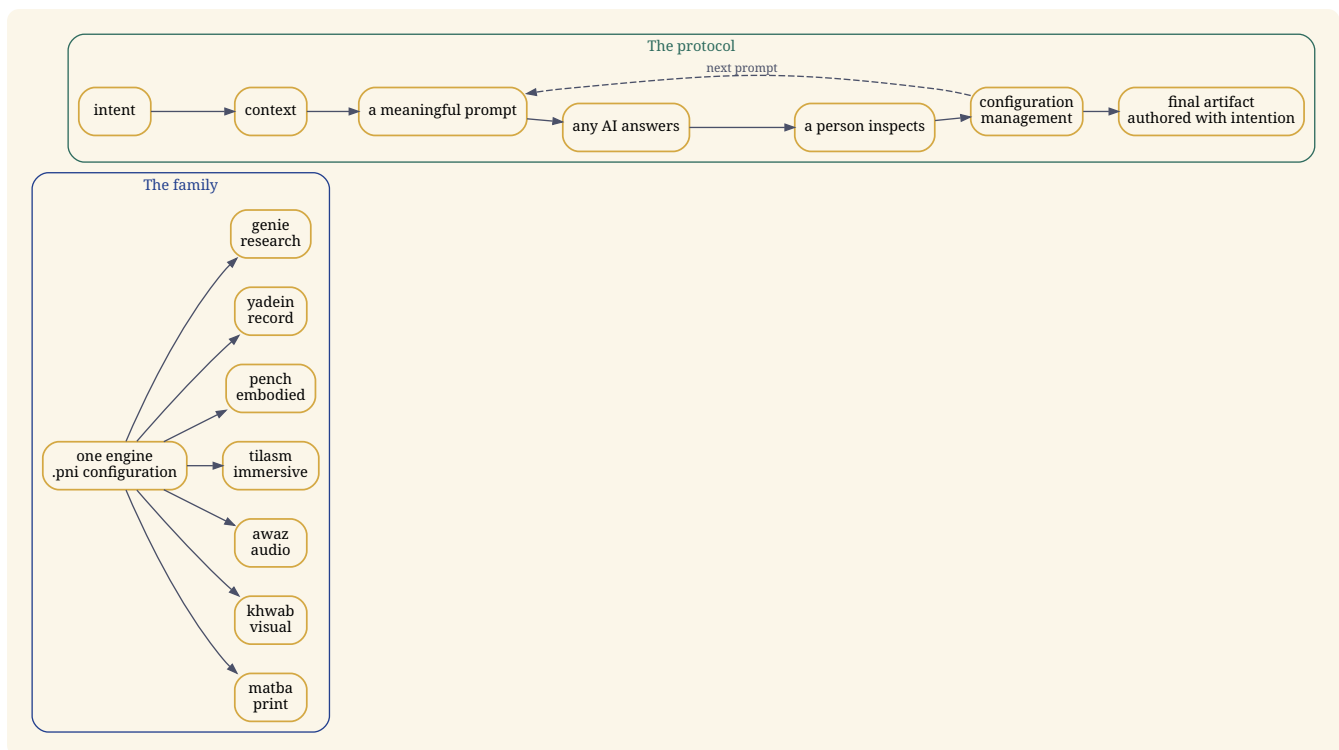


Figure 5. The cyclers: one engine, configured in PANINI, and the protocol every cycle follows.

Provenance, in a fixed order

Four systems, never collapsed into one. **Tok DOI** proves that a thing existed by a given time. **spiguard** is the disclosure gate, which fails closed. **Candor** records signed intent as in-toto statements. **Misty DOI** is the one that reaches the world, through Zenodo. The order is always seal, clear, attest, mint. Chapter 25 explains the machinery.

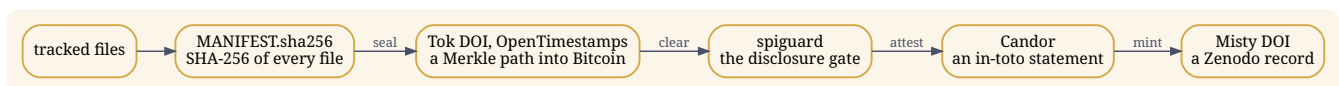


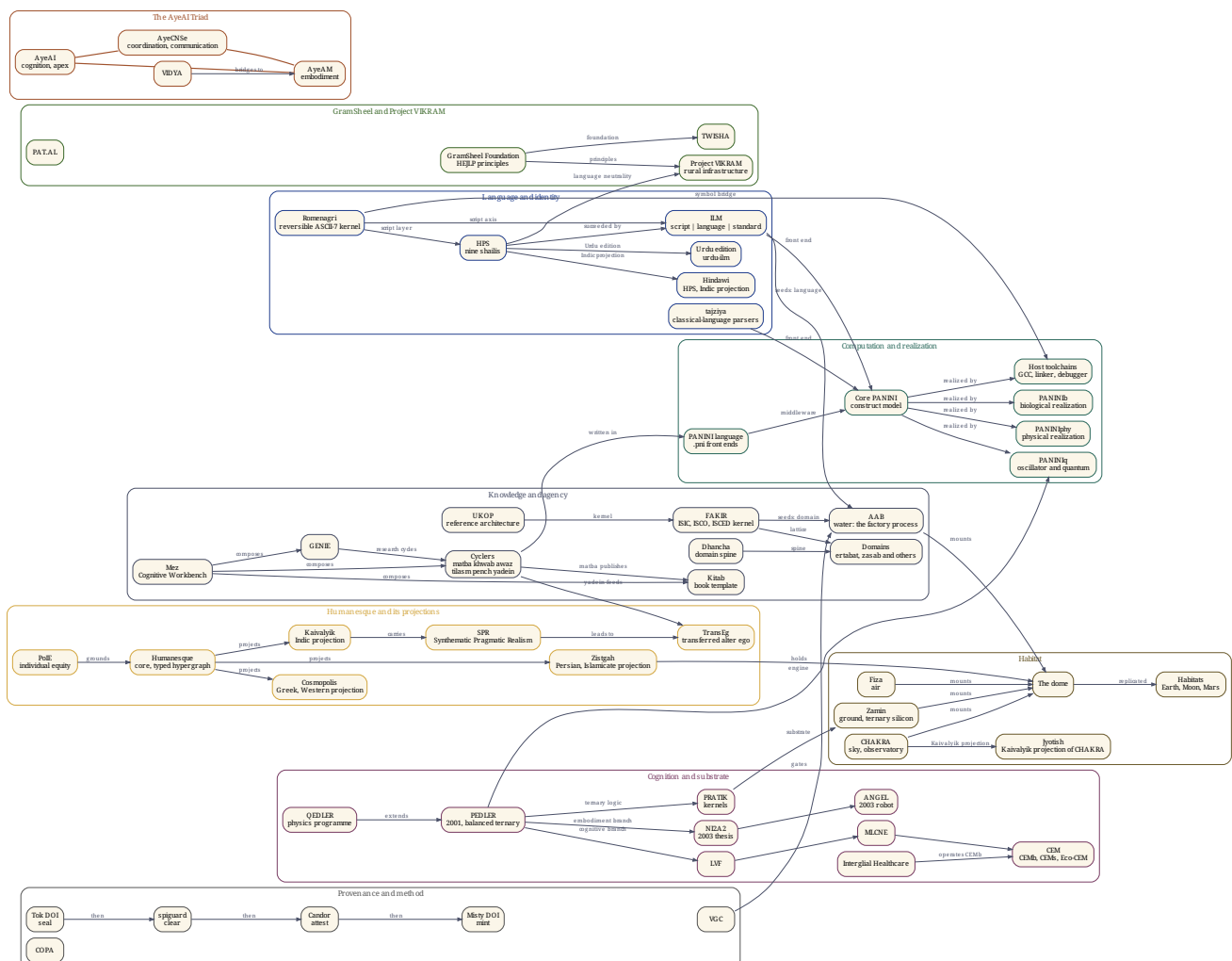
Figure 6. Provenance in its fixed order: seal, clear, attest, mint.

7. The dependency graph

In plain words. A map of how the parts connect, like a city map with roads between places. Any route you follow is one journey; the map is the whole picture.

The estate is a typed graph, and any account of it in words is one traversal through that graph. Read a traversal, then return to the graph: adjacency is not dependency, a projection is not a hierarchy, composition is not absorption, and an Act is not a stage of maturity.

Every node and every edge below is taken from the retrieved record: the repositories, their READMEs and the author's statements. Clusters group the nodes by family, not by layer, and an edge label names the relation it records. The graph's source is `data/graph.json`; `python3 tools/graph.py` renders it, and the build keeps the rendered `dependency-graph.svg` and its Graphviz source `dependency-graph.dot` beside this guide.



Three traversals

A name, from a Devanagari source to a debugger. Romenagri is the script layer of HPS and the script axis of ILM; through the symbol bridge it carries every name into the host toolchain, and the inverse brings it back (Chapters 19 and 21).

A task, from a domain and a language to a sealed artifact. FAKIR seeds the domain and ILM the language; AAB builds; the cyclers, written in the PANINI language and composed in Mez, produce the work; matba publishes it through Kitab; Tok DOI, spiguard, Candor and Misty DOI seal, clear, attest and mint it, in that order; VGC gates the work between people and AI throughout.

A mind, from an event to a substrate. PEDLER's cognitive branch runs through LVF and MLCNE to the cognitive enablement modules, which Interglial Healthcare operates on the biological side; its embodiment branch runs through NI2A2 to ANGEL; its balanced-ternary logic runs into PRATIK and down to Zamin, the ground of the dome; and its engine drives PANINIq.

8. The ecosystem, term by term

In plain words. A dictionary of the special words used here, each with the nearest everyday idea and what makes it different.

Each entry gives what the thing is in this architecture, the nearest analogue a newcomer is likely to reach for, and the exact difference. The analogue is a door into the idea, never a definition of it. The state in brackets says what exists: established (historical record), released (on the forge, with its DOI where minted), executed (run for this guide), under construction, specification, or planned. Entries marked † follow the author's working notes and write-ups as supplied on 29 September and 3 October 2026, rather than the repositories.

Language and identity

ILM, Integrative Linguistic Multiscript (released; ilm.codes live). *In the architecture:* the linguistic pillar and the identity layer for language, with script, language and standard kept as separate axes, and layers for phonology, transliteration, orthography, lexicon, syntax and semantics, and interfaces, supported by data, validation, development tools, language specifications and a language server; its registry lists 7,867 languages and 226 scripts. *Nearest analogue:* localisation, an NLP toolkit, a transliteration library. *Difference:* it keeps the three axes apart, so that any language can travel in any script, and it gives every script a reversible path through the whole computing stack, down to the symbol table.

Romenagri (established 2003; released; executed). *In the architecture:* the reversible kernel between scripts and ASCII-7. Its paper states six constraints met together: ASCII-7 closure, case independence, diacritic independence, legality as a C identifier, reversibility and linear time, over the Brahmi, Perso-Arabic and Northwest Semitic families, with a canonical-form layer and an ASCII-reduction layer. *Nearest analogue:* romanisation, such as IAST, ISO 15919 or ITRANS. *Difference:* its output is a valid identifier for every tool from lexer to debugger, and it inverts, so a name survives compilation and comes back.

HPS, the Hindawi Programming System (established 15 August 2004; released; executed). *In the architecture:* nine shailis over nine host languages, from assembly, lex and yacc to C++, Java, Python, BASIC and LOGO; a script layer, a language layer and the host standard, composed as transducers over unmodified host toolchains; now HPS's Indic projection under ILM, with an Urdu edition beside it. *Nearest analogue:* a localised programming language. *Difference:* a complete systems stack, not a keyword translation, with names carried through Romenagri.

tajziya (released, doi:10.5281/zenodo.22982238). *In the architecture:* the parser federation for the classical languages, by family, with the author's Sanskrit parser; the front end of the PANINI stack beside ILM. *Nearest analogue:* NLP parsers. *Difference:* organised to feed PANINI's construct model and AAB's paintings.

Computation

PANINI (several members released; some under construction). *In the architecture*: the construct model at the core, with ILM and tajziya in front, PANINI's own front ends and language as middleware, and realization backends behind; it runs through all three Acts. Its family includes the prompt-cycle language in which the cyclers are written (zistgah/panini), the general-purpose PANINI of the merged release (zistgah/humanesque), and the realization arms. *Nearest analogue*: a compiler, a programming language, an intermediate representation. *Difference*: programming languages, prompt cycles and physical, biological and quantum realizations are all projections of one construct model.

PANINIQ (released, doi:10.5281/zenodo.23020648; executed). *In the architecture*: a realization backend: an oscillator substrate, the PEDLER engine and quantum validation. *Nearest analogue*: a quantum simulator. *Difference*: the substrate is an oscillator Ising machine with a learning automaton on it; the quantum circuit validates it rather than replacing it.

PANINIB and **PANINIPHY** (released research platforms, version 1.0). *In the architecture*: PANINIB, the biological realization arm, lowers biological intent through semantic, resolution, CISC, RISC and sequence IRs to a realization plan and an execution run; PANINIPHY, the physical realization arm, lowers physical intent through a physical IR and physical-domain engines, down to modular blocks and the selection of parts by force, stroke and voltage. *Nearest analogue*: computer-aided design, genome design tools. *Difference*: both share PANINI's common spine, language, verification, provenance and ArtifactGraph, adding only their own domain IR and engines; laboratories and machines are external adapters, and neither claims a result its evidence does not show.

Cognition

PEDLER and **QEDLER** (established November 2001; record doi:10.5281/zenodo.17497559; QEDLER a research framework). *In the architecture*: the point-event-driven learner; a six-tuple extension of the Turing machine over balanced ternary; the source of the primitives Act (executable intent), Inclination (a signed directional gradient on transition edges) and Natural Justice (a gate on the passage from intent to Act); QEDLER extends it to physics. *Nearest analogue*: an event-driven or neuromorphic learning model. *Difference*: it predates the present generation of models and carries the executable-intent semantics the constitutional corpus is built from.

PRATIK and **Zamin** (released: pratik_core_mvp doi:10.5281/zenodo.21288232, zamin doi:10.5281/zenodo.21297556). *In the architecture*: PRATIK, Participatory Recursive Adaptive Trans-Intelligence Kernels, on the divider between Acts I and II, with a kernel in C++ and CUDA; Zamin, its physical balanced-ternary substrate with a poised zero. *Nearest analogue*: a neuromorphic kernel on novel hardware. *Difference*: balanced ternary and event-driven from the logic up.

CEM, with CEMb, CEMs and Eco-CEM (specification; Act II). *In the architecture*: cognitive enablement modules over one kernel, $\mathcal{E} = (\mathbf{E}, \mathbf{C}, \mathbf{\Pi}, \mathbf{W})$, realised on biological substrates (BCI, prosthetics, closed-loop cognition, operated through Interglial Healthcare), on synthetic ones (processors, distributed systems, robotics) and on ecological ones. *Nearest analogue*: robotics, BCI, prosthetics. *Difference*: one invariant kernel across substrates; the superscript is part of the name.

The AyeAI Triad (specification with formal closure). *In the architecture*: AyeAM = $\langle \mathbf{S}, \mathbf{R}, \mathbf{C} \rangle$ for embodiment, AyeAI = $\langle \mathbf{M}, \mathbf{I}, \mathbf{G} \rangle$ for cognition, AyeCNSe = $\langle \mathbf{T}, \mathbf{Ch}, \mathbf{\Sigma} \rangle$ for coordination and communication across media, with AyeAI at the apex. *Nearest analogue*: a perception, cognition and

action loop. *Difference*: a triad, not a loop and not a pipeline.

Knowledge and agency

UKOP, FAKIR and Dhancha (FAKIR released, doi:10.5281/zenodo.21436550; Dhancha released, doi:10.5281/zenodo.22821645; UKOP a specification). *In the architecture*: UKOP, the reference architecture for human knowledge; FAKIR, its kernel over ISIC, ISCO and ISCED crossed with AGI layers and language; Dhancha, the spine every domain is built on, whose rule is that the engine is common and the workflow is not. *Nearest analogue*: a knowledge graph or reasoning engine. *Difference*: domain invariants are enforced by tests, and resolution and verification are primitives.

Cyclers (six released and minted; Act I). *In the architecture*: matba, khwab, awaz, tilasm, pench and yadein, classified by what they produce, plus genie, each with its own contract, state model and evidence rules, written in PANINI and run by one engine. *Nearest analogue*: an agentic harness or orchestration loop. *Difference*: AI-agnostic by construction, inspected by a person at every step, and the recorded method, never the content, is itself the reproducible work. † An older form of the corpus names SAFAR in the sixth position, where the current corpus has yadein.

Kitab (released). *In the architecture*: the config-driven book template that matba publishes through, and † in the author's notes the persistent artifact and publication layer that keeps an artifact's state, its creative lineage, its execution trace and evaluations, its forks and descendants. *Nearest analogue*: a publishing template. *Difference*: a cycler publishes through Kitab; Kitab is not a seventh cycler.

Research Kundali (released). *In the architecture*: a research map of a person, a lab or a paper, built from an ORCID, a lab name or a DOI ([project-ilm/research-kundali](https://project-ilm.com/research-kundali)); † in the author's notes, the systematic record of claims, artifacts, chronology, provenance, objections, evidence and responses, so that every claim can be checked against its evidence. *Nearest analogue*: a researcher profile. *Difference*: it records evidence and discrepancies, not reputation.

GENIE (released). *In the architecture*: the Generalized Emotive-Narrative Interaction Engine, storyteller, poet and painter, and the research cycle through its Prompt Operating System, with six verbs (create, verify, execute, measure, falsify, integrate), nine epistemic tags and a gate that a simulation cannot pass in place of an experiment. † In the author's own formulation GENIE is itself a cycler, the creative cycler, which can decorate the modality cyclers without erasing their identities. *Nearest analogue*: a creative or research AI agent. *Difference*: governed by a constitution of primitives; orchestration and composition, not an implementation.

AAB (released; Act I). *In the architecture*: the paint program for systems, from painted intent to verified, sealed artifact; the gamified studio of verification-gated quests; home of the process cyclers; the process half of the estate's software factory, with FAKIR as its component registry. In the author's words, AAB and FAKIR together constitute the working definition of AGI used here. *Nearest analogue*: a low-code studio or a software factory. *Difference*: every task is a painting with a manifest, quest stages and oracle-gated verification, and every component keeps its provenance.

Mez, the Cognitive Workbench (released). *In the architecture*: the local-first desk that brings independently existing systems together for an exercise: composition, exercise, observation, synthesis, a new artifact. *Nearest analogue*: an IDE or a notebook. *Difference*: composition without absorption; every composed system keeps its own identity and life.

Method and evidence

VGC, Verification-Gated Human-AI Co-Development (released, doi:10.5281/zenodo.21264248). *In the architecture*: the method by which people and AI build together, with every step gated by verification. *Nearest analogue*: test-driven development or human review of AI output. *Difference*: the gate is a verified artifact, not a reviewer's impression, and the AI's report that something passed is never the evidence that it passed.

COPA, the Cost of Perceived Authority (released, doi:10.5281/zenodo.21782217). *In the architecture*: a protocol for measuring how much an AI system asserts before it verifies, through the Authority Projection Index, with its hypothesis stated beside a fair null. *Nearest analogue*: an AI evaluation benchmark. *Difference*: it measures projected authority rather than accuracy, and its target is calibrated trust.

† One rule binds all of these: a record that has not been retrieved is not thereby false. Unretrieved is a state of the reader's context, not of the world, and a claim moves from unknown to false only on evidence.

Continuity and civilisation

Humanesque (specification; the merged release on the forge). *In the architecture*: the core over which everything above is a typed hypergraph, and the constitutional realm derived from PoIE through recursive sovereignty. *Nearest analogue*: an AGI architecture. *Difference*: a constitutional and epistemic realm as well as a technical substrate, not a product.

Kaivalyik, Zistgah and Cosmopolis (projections). As in Chapter 6: parallel cultural projections of the whole, not stages. Kaivalyik carries Synthematic Pragmatic Realism, SPR, the framework that heads Act III ("Kaivalyik: Towards a Synthematic Pragmatic Realism for the AGI Singularity"); Zistgah carries the habitats and the elements.

TransEg (released, doi:10.5281/zenodo.21321558). *In the architecture*: the transferred alter ego, the mechanism by which a constituted identity continues; a local digital-twin reference implementation; yadein's staggered upload feeds it. *Nearest analogue*: a digital avatar. *Difference*: continuity of an identity under the constitutional corpus, not a persona.

PAT.AL, VIDYA, GramSheel, Project VIKRAM and TWISHA (established). PAT.AL, the Participatory Alliance for Technology, Access and Livelihoods, is participatory infrastructure; VIDYA bridges to AyeAM; Project VIKRAM, Virtualized Infrastructure for Knowledge-driven Rural Ascension Management, runs under the GramSheel Foundation beside TWISHA. They sit on the dividers between the Acts. The question the author puts to them is whether technology can distribute capability without concentrating power.

The reference lab (planned; its launch scripts are the next release). An AGI-capable laboratory for 30 to 40 thousand, able to carry 80 to 90 per cent of the research behind high-end AI, robotics and automation papers, repeated as a pattern for every domain and aligned with GATE-level courses in computer science and in robotics and automation.

Also in the working notes

Modular self-reconfiguring robots †: very simple modular blocks with face-to-face connections and electro-permanent-magnet contacts, assembling into larger structures, ruggedised for decentralised manufacture with low-cost embedded control; PANINIphy's modular-block examples are the compiler side of this. **Further realization families** †: chemical, photonic, and a computational-biological intermediate layer of the AlphaFold class, built as an independent spine rather than a dependency. **The build engine** †: the direction of the whole is a self-reflecting, self-evolving and self-sustaining build engine in which PANINI captures and expresses the construct and the intent, and agency, human or machine, biological or physical, does the building. **INDUS and KHĀK** † are named in the notes among the lineages and the robots, alongside ANGEL, TARA and NI2A2.

9. The three Acts

In plain words. Three stages of one long story: machines that think with us; machines that help us build with living, synthetic and natural materials; and people and their worlds carrying on into the future.

The author's note "3 Act ASI $\wedge$ Panini" and his dictation of 27 September 2026 place the components in time. An Act says when a piece becomes most consequential, not when work on it began: much of Act III is being worked on now. The Acts are a temporal coordinate; the projections of Chapter 6 are cultural ones; neither is a ladder of maturity.

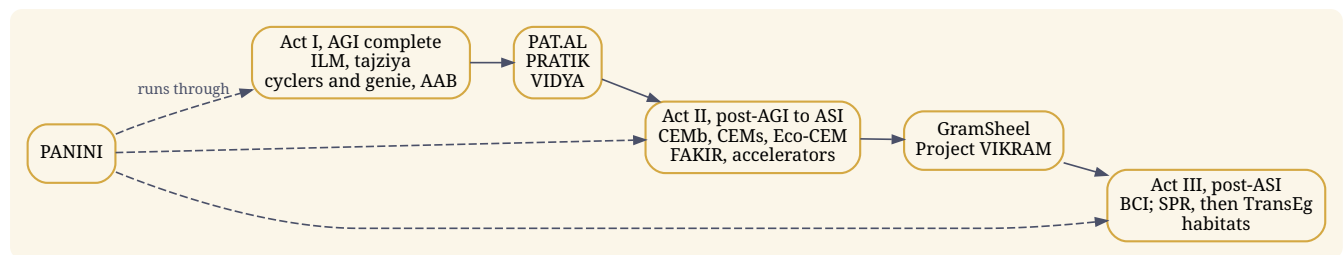


Figure 7. The three Acts, with their dividers; PANINI runs through all three.

Act	What it holds
Act I, AGI complete	Sanskrit and Pāṇini, with cyclers and paradigms; the front end, ILM and tajziya; the cyclers named by output with genie, which in market terms are agents and harnesses; the process cyclers, designed and run in AAB
Divider between I and II	PAT.AL, PRATIK, VIDYA
Act II, post-AGI to ASI	The cognitive enablement modules, CEMb and CEMs, plus Eco-CEM, with FAKIR; the accelerators
Divider between II and III	GramSheel and Project VIKRAM
Act III, post-ASI	All the BCI work; Synthematic Pragmatic Realism, then TransEg, the transferred alter ego; the habitats: Humanesque, with Zistgah on Earth (Asli Zistgah), the Moon (Zistgah-e-Mahtab) and Mars (Zistgah-e-Bahram), Kaivalyik, Cosmopolis, and more left open
Across all three	PANINI
Beyond the Acts	The reference lab and the aligned courses

Act I: intelligence that works through language

The machinery of Act I is the front end (ILM and tajziya), the cyclers with genie, and AAB, with PANINI as the notation of the cycles. † In the author's notes, Act I is complete when intelligence can take an intent through language, in any language and script, into executable specifications, run it through cycles a person inspects, research and create with it, and keep the provenance of what it makes. On the first divider stand PAT.AL, the participatory infrastructure, PRATIK, the kernels, and VIDYA, the bridge to AyeAM.

Act II: from agency to realization

† In the author's notes, Act II is where the architecture moves from cognitive and digital agency to cognitive enablement and realization on substrates. The cognitive enablement modules carry one kernel to biological, synthetic and ecological substrates; FAKIR supplies the knowledge they act on; the accelerators carry the load. The notes tie this Act to realization on substrates, which PANINI's realization arms carry out: PANINib for biology and PANINIphy for physical systems, each lowering intent through the common spine of Chapter 6 and each keeping desired, predicted, observed and validated results apart. † In the author's notes the same spine runs intent, specification, knowledge resolution, construct, realization selection, planning, geometry and trajectory, execution, observation, evidence, verification and validation, provenance and a knowledge update, and the states of a result are kept apart all the way: desired, designed, predicted, planned, realized, executed, observed, verified, validated. On the second divider stand GramSheel and Project VIKRAM: the foundation with its principles, and the rural digitalisation platform inspired by them.

Act III: continuity

Act III gathers the brain-computer interface work (Quantum Neuromorphic BCI, brain-robot interfaces and NeuroMusical Therapeutics as active research; brain-to-text, cognitive prosthetics and intelligent rehabilitation as research directions needing clinical partners), Synthematic Pragmatic Realism, and TransEg, through which a constituted identity continues beyond the substrate that first carried it. Its habitats are Humanesque itself and its projections, with Zistgah's domes on Earth, the Moon and Mars; the Moon carries a lander embodiment and Mars a quadrotor one, and both replicate the Earth dome.

PANINI runs through all three Acts.

10. Three families, set apart: the AyeAI Triad, GramSheel and Project VIKRAM, Humanesque and its projections

In plain words. Three groups explained one at a time: the AyeAI Triad of thinking, body and communication; GramSheel and Project VIKRAM for villages; and Humanesque with the names it takes in each culture.

Three families of work are easy to blur into each other or into the rest of the estate. This chapter sets each apart, with what its own record says.

The AyeAI Triad

The Triad has three members, AyeAI®, AyeAM® and AyeCNSe®, with AyeAI at the apex and AyeCNSe and AyeAM at the base. It is a triad, not a pipeline and not a loop: a loop running from the environment through AyeCNSe, AyeAI and AyeAM back to the environment was proposed elsewhere and rejected. Its formal closure gives each member a tuple:

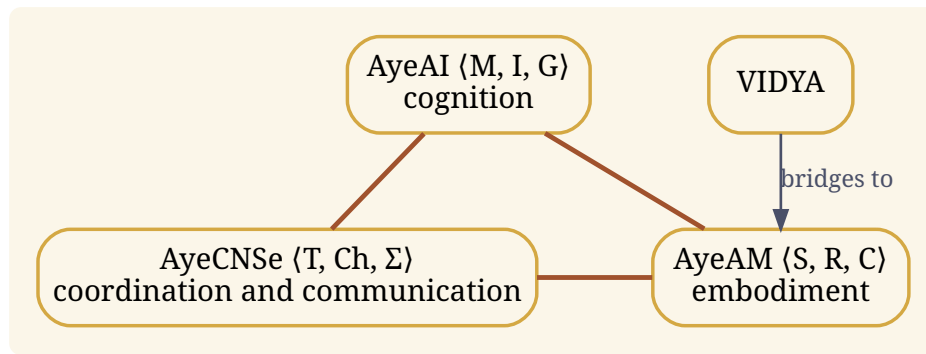


Figure 8. The AyeAI Triad: a triad with AyeAI at the apex, not a loop and not a pipeline.

$$\mathbf{AyeAM} = \langle S, R, C \rangle, \quad \mathbf{AyeAI} = \langle M, I, G \rangle, \quad \mathbf{AyeCNSe} = \langle T, Ch, \Sigma \rangle.$$

AyeAI is cognition, the apex: “the inclusive AI”, in the words of its site, and the organisation [ayelai](#) holds its platforms, among them an open public stack for synthetic intelligence (opssi), virtual deployment infrastructure for scientific and cognitive computing (ayevdi), Staggered Upload™ (upload), the AyeAI Singularity Public License (spl) and the NI2A2 work.

AyeAM is embodiment: the body through which intelligence acts, across ground, air, water, space and habitat. VIDYA bridges to AyeAM without being a member of the Triad.

AyeCNSe is coordination and communication across media: the form studio at [forms.ayecnse.site](#) and the Human Context Model paper at [hcm.ayecnse.site](#) are its published faces.

Two repositories mark themselves as PEDLER and AyeAI Triad work: PRATIK’s kernel and Zamin’s ternary silicon.

GramSheel and Project VIKRAM

The GramSheel Foundation is the social architecture of the estate and the older of its lines: † it began as the Nature Welfare Society in 1993. Its principles, HEJLP, are health, education, justice, livelihood and peace. Its Village Knowledge Center (Universal Device) was a Stockholm Challenge finalist in 2010, and TWISHA runs under the same foundation.

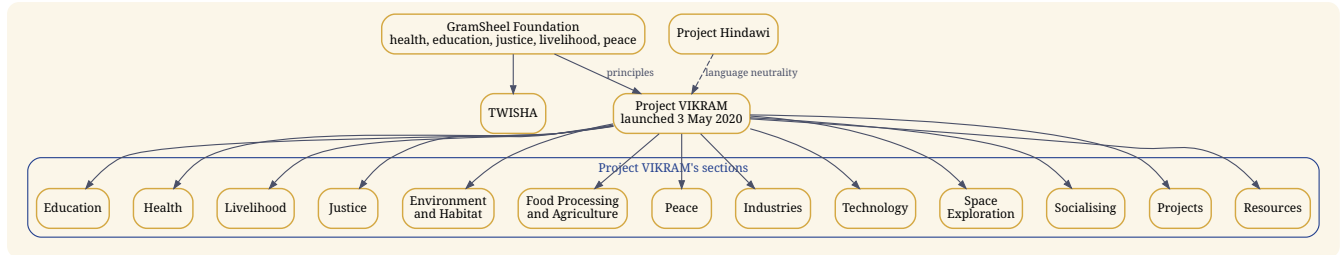


Figure 9. GramSheel and Project VIKRAM, with VIKRAM's sections as its own pages name them.

Project VIKRAM, Virtualized Infrastructure for Knowledge-driven Rural Ascension Management, is † the technical implementation of GramSheel's principles. It was launched on 3 May 2020 as the Project Vikram Journal, “an inclusive rural digitalization platform inspired by the principles of GramSheel”, under the motto गाँव बढ़ेंगे तो सब बढ़ेंगे, when the villages grow, everyone grows. Its own pages (pvjournal.github.io) name its components:

Component	What its page covers
Education	learning and schooling
Health	healthcare, open to submissions as issues, with tools for an accessible virtual hospital
Livelihood	the ISIC domains of economic activity
Justice	social justice
Environment and Habitat	the living environment
Food Processing and Agriculture	agriculture and food
Peace	harmony through social dialogue
Industries, Technology and Space Exploration	the technical frontiers it reaches towards
Socialising, Projects and Resources	the community and its shared material

With Project Hindawi, its page says, VIKRAM works towards language neutrality across technical fields. In the three Acts, GramSheel and Project VIKRAM stand on the divider between Acts II and III.

Gram Swaraj and the questions behind GramSheel and VIKRAM

Gandhi's *Hind Swaraj* (1909) argued that self-rule is more than a transfer of authority, and his *Constructive Programme: Its Meaning and Place* (1941) placed the rebuilding of daily life inside the work of independence: its eighteen items include village industries, village sanitation, basic and adult education, the provincial languages and a national language, economic equality, the kisans, labour, the adivasis and women. The estate does not present itself as a restatement of these ideas. The point of contact is a question that becomes unavoidable once computation is infrastructure: can a community govern itself if the knowledge and the machines its decisions depend on are controlled elsewhere?

On Gandhi Jayanti, 2 October 2026, the author put the questions behind GramSheel, Project VIKRAM and Humanesque in his own words. Among them:

- Can rural communities become producers of knowledge and technology, not merely consumers of systems designed elsewhere?
- Can technology distribute capability without concentrating power?
- Can artificial intelligence strengthen individual and community sovereignty rather than replace it?
- Can knowledge remain plural, multilingual and culturally grounded while becoming computationally executable?
- Can verification, provenance and human agency become constitutional properties of intelligent systems?
- Can GramSheel build the local knowledge infrastructure, can Project VIKRAM turn that infrastructure into practical capability, and can Humanesque provide a larger architecture in which intelligence, identity, agency, provenance and equity remain connected?

His answer to the old division of the world is a direction rather than a side: to march together, not as a Global South or a Global North, but towards a Global Centre, where capability is distributed and humanity remains sovereign.

Humanesque and its projections, with their names

Humanesque is the core, with its site at humanesque.site and its organisation at hmnsq: the shared executable typed hypergraph, the ontology federation, the common execution and evidence spine, and the components every projection reuses. Its constitutional ground is PoIE, from which sovereignty recurses to every level. Its core stem repositories all belong to Humanesque; each projection gives them its own culturally sensitive names and sites.

Kaivalyik, the Indic projection, including the Advaita framing (kaivalyik.org, organisation [kaivalyik](https://kaivalyik.org)): consciousness, intelligence and transcendence; it carries Synthemetic Pragmatic Realism, Kaivalyik AGI and Jyotish. **Zistgah**, زستگاه, the Persian and Islamicate projection (zistgah.org, organisation [zistgah](https://zistgah.org)): life, planet and ecology, a habitat to live and work in; it carries the elements, the dome and the habitats on Earth (Asli Zistgah), the Moon (Zistgah-e-Mahtab) and Mars (Zistgah-e-Bahram, † conceived earlier as Mangal Kamna). **Cosmopolis**, the Greek, Western and classical projection (cosmopolis.org, organisation c-polis): societies, civilisations, space and deep time.

The stem names below are the projection-neutral names of the components. Today the working repositories live in the zistgah organisation, so most of them carry their Zistgah names; two stems already show a second projection, HPS in Hindawi and CHAKRA in Jyotish.

Stem, in Humanesque	Zistgah	Kaivalyik
The programming system	Urdu edition (urdu-ilm)	Hindawi
The observatory and its calendars	Chakra, the turning sky	Jyotish
The ground: balanced-ternary silicon	Zamin, زمین	
The water: the factory process	AAB, آب	
The air: the environmental replica	Fiza, فضا	
The shared virtual dome	the dome	
The Cognitive Workbench	Mez, میز	
The press	Matba, مطبع	
The cutting room, the visual cyler	Khwab, خواب	
The listening room, the audio cyler	Awaz, آواز	
The immersive cyler	Tilasm, طلسم	
The embodied cyler	Pench	
The record	Yadein, یادیں	
The book template	Kitab, کتاب	
The domain spine	Dhancha, ڈھانچہ	
Communication engineering	Ertabat, ارتباط	
Device control	Zasab, اعصاب	
The door for any AI	Darwaza, دروازہ	
Scan and store	Tasvir, تصویر	
The plates	Lawh, لوح	
The parser federation	Tajziya, تجزیہ	
The guide	Rahnuma, رہنما	
The Geometry of Becoming	Tabdili, تبدیلی	
The Geometry of Living	Zindagi, زندگی	
The Geometry of Persisting	Qaem, قائم	
Topological Bildung	Tarbiyat, تربیت	

Some stems keep one name in every projection: Humanesque, ILM, Romenagri, PANINI with PANINIq, PANINIb and PANINIphy, PEDLER, QEDLER, PRATIK, CEM, UKOP, FAKIR, GENIE, TransEg, PoIE, VGC, COPA, and the provenance systems Tok DOI, spiguard, Candor and Misty DOI. Kaivalyik's and Cosmopolis's own names for the rest are given with their organisations, as each projection takes its repositories.

Appendices

B. Reading the architecture without flattening it

In plain words. How to read the architecture without squashing it into a straight line: what an arrow can mean, and the loops that run at different scales.

The architecture is dense because what it addresses is dense: language, cognition, knowledge, participation, embodiment, verification, provenance, community capability, identity and continuity. This appendix does not simplify it by removing parts. It shows how to read it so that its structure stays open to questioning. Statements marked † follow the author's working notes and write-ups as supplied on 29 September and 3 October 2026.

A graph read one route at a time

Any sequence printed on a page, A then B then C, is one route through the graph of Chapter 7, chosen because a sequence is easier to read. Before trusting an arrow, ask which of seven things it means:

The arrow is	Example in this guide
temporal: one thing came before another	the dated record; PEDLER (2001) before Romenagri (2003)
causal: one thing brings about another	an observation changing a knowledge state
a dependency: one cannot be built or understood without the other	the cyclers are written in the PANINI language
an interface: two things meet at a defined boundary	Romenagri names crossing into the host toolchain
a composition: things used together without merging	Mez composing GENIE, Kitab and the cyclers
an expository order: the order of explanation only	the reading paths of the reader views
an interpretive mapping: a reader's bridge between vocabularies	the nearest analogue in each term entry

These are not interchangeable. The estate itself keeps three of its orderings apart: the Acts are temporal, the projections are cultural, and the edges of the graph are architectural; none of them is a ladder of maturity.

Loops at different scales

The architecture has several loops, and they run at different scales; collapsing them into one “AI feedback loop” loses what each is for.

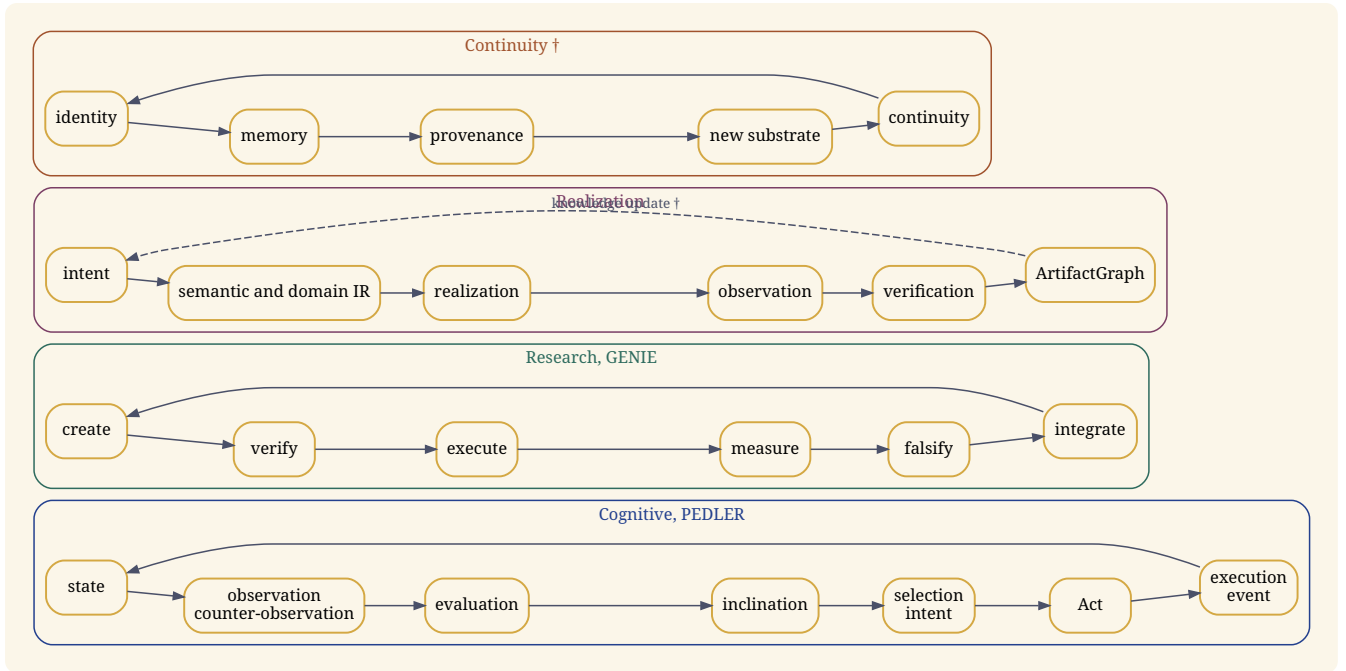


Figure 14. Loops at four scales: PEDLER's, GENIE's, the realization spine's, and continuity as the write-ups read it (†).

The cognitive loop, PEDLER's. State, observation, counter-observation, evaluation, inclination, selection, intent, Act, execution, event, and a new state. Inclination is a signed directional gradient, not yet an intention; the Act is executable intent; Natural Justice gates the passage from intent to Act.

The research loop, GENIE's. Create, verify, execute, measure, falsify, integrate, with nine epistemic tags on every statement and a gate that a simulation cannot pass in place of an experiment.

The realization loop, PANINI's. Language, semantic IR, realization requirement, domain resolution, domain IR, then a common layer of geometry, trajectory, animation, observation, verification and an ArtifactGraph. The realization repositories keep desired, predicted, observed and validated results apart, and compilation success, execution success, target satisfaction and validation apart; † the notes extend the separation to desired, designed, predicted, planned, realized, executed, observed, verified and validated.

The continuity loop. † The write-ups read a fourth loop at civilisational scale: identity, memory, provenance, transformation, a new substrate or embodiment, continuity. Its components in the record are yadein, which records, TransEg, which continues a constituted identity, and the provenance systems that keep what happened.

In every loop the gates do work in both directions. Verification is not only a last inspection: a failed gate stops the next step. Provenance is not only storage: it fixes what can legitimately be claimed about an artifact. Observation is not only telemetry: it changes the state from which the next action is chosen. VGC states the rule plainly: the gate is a verified artifact, and an AI's report that something passed is not evidence that it passed.

One component, several cycles

A component is not exhausted by the place it appears in one diagram. ILM belongs to the linguistic line, yet language shapes how knowledge and intent are represented. FAKIR is a knowledge kernel, yet the knowledge it supplies shapes what is done. PEDLER is a cognitive architecture, yet its Act, Inclination and Natural Justice are also the primitives of the constitutional corpus. VGC concerns verification, yet verification decides what may become accepted knowledge. GramSheel is a foundation for villages, yet it touches education, livelihood, health and technology at once. The graph records relations of dependence, interaction and constraint, not modules stacked in a tower.

Names are interfaces, and analogies are doorways

The density of names cannot be removed without removing distinctions; each name denotes a different object. What helps is a functional handle at first meeting, which the term-by-term chapter and Appendix A give, and an analogy used as a doorway rather than a definition. † In the write-ups' phrase, the analogy opens the door; the formal definition decides what is inside the room.

F. Formal definitions

In plain words. The same ideas written as mathematics, for readers who want the formal version.

This appendix states in symbols what the chapters state in words. Where the record gives a formal object, the PEDLER six-tuple, the CEM kernel or the Triad's tuples, it is reproduced as recorded. Where the record gives only words, the formalisation is this guide's reading, and is said to be.

The estate as a typed graph

The estate is a typed graph

$$H = (V, E, \tau_V, \tau_E), \quad E \subseteq V \times V \times R,$$

where V is the set of components, R the set of relations (projects, script layer, realized by, front end, mounts and the rest), $\tau_V : V \rightarrow \mathcal{F}$ assigns each component its family, and τ_E each edge its relation. The core is described as a typed hypergraph, in which one edge may join several components at once: $E \subseteq 2^V \times R$. A traversal is a walk $v_0 \xrightarrow{r_1} v_1 \xrightarrow{r_2} \dots \xrightarrow{r_k} v_k$; every sequence printed in this guide, and every reading path of the reader views, is one.

In this guide's reading, a projection $p \in \{\text{Kaivalyik, Zistgah, Cosmopolis}\}$ is a map $\pi_p : H \rightarrow H_p$ that keeps every component's stem and every relation and changes names: $\text{stem}(\pi_p(v)) = \text{stem}(v)$ and $(u, v, r) \in E \Rightarrow (\pi_p(u), \pi_p(v), r) \in E_p$. The Acts are a separate, partial labelling $\alpha : V \rightarrow \{\text{I, I|II, II, II|III, III}\}$, with PANINI in every Act.

Romenagri as a code

Let Σ be the Devanagari alphabet and $\Gamma = \{\mathbf{a}, \dots, \mathbf{z}, _ \}$, a subset of the alphabet of C identifiers. Romenagri's compiler form is a map

$$T : W \rightarrow \Gamma^*, \quad W \subseteq \Sigma^*,$$

injective on well-formed text W , with an inverse T^{-1} on $T(W)$. Text outside W is first brought to its canonical form by $c : \Sigma^* \rightarrow W$, so that for every string

$$T^{-1}(T(s)) = c(s), \quad c(s) = s \text{ whenever } s \in W.$$

On the 984 words of the Hindi corpus, $T^{-1}T(w) = w$ for 983; the one exception is a misspelling in the corpus.

For a Brahmi script σ whose Unicode block follows the ISCII layout from base b_σ , the hub map is $h_\sigma(x) = x - b_\sigma + \mathbf{0x900}$ on the aligned letters, a bijection onto the corresponding Devanagari letters; `flatten_uni_dev` applies it by table. A word in script σ therefore has the identity $T(h_\sigma(w))$, and returns by $h_\sigma^{-1}(T^{-1}(\cdot))$.

Urdu meets the hub through $\varphi : \Sigma_{\text{ur}}^* \rightarrow \Sigma^*$, which is many-to-one on unmarked text: the fibre $\varphi^{-1}(w)$ over a hub word can hold several spoken readings, because the short vowels are not written. With tashkil, the restriction of φ to marked text distinguishes them: unmarked ہندوی maps to हन्दवी , and marked ہندوی to हिन्दवी . In the corpus, 1 word in 794 carries a vowel mark.

The symbol bridge

For every name n in a program, the toolchain sees $\text{sym}(n) = T(n) \in \Gamma^*$. Each stage t_k between source and debugger (compiler; assembler; linker, ELF writer, DWARF writer; debugger) carries identifiers over Γ unchanged: $t_k(\text{sym}(n)) = \text{sym}(n)$. Hence

$$T^{-1}(t_m \circ \dots \circ t_1(\text{sym}(n))) = c(n),$$

which is the name as written. Chapter 21 runs this for every name of a program.

The construct model

Let $\mathcal{C}$ be the constructs (39), $\mathcal{L}$ the languages (27) and $\mathcal{H}$ the host languages (8). Each language ℓ has a keyword map $\kappa_\ell : \mathcal{C} \rightarrow W_\ell$, and each host h a realisation $\rho_h : \mathcal{C} \rightarrow W_h$, partial because not every host has every construct. Translation from ℓ to h is

$$\tau_{\ell \rightarrow h} = \rho_h \circ \kappa_\ell^{-1},$$

defined when κ_ℓ is injective, which is the reversibility check ILM runs on every keyword table. Direct pairings of a language with a host would number $|\mathcal{L}| \cdot |\mathcal{H}| = 27 \times 8 = 216$ tables; the construct model needs $|\mathcal{L}| + |\mathcal{H}| = 35$.

PEDLER, as PANINIq implements it

PANINIq implements PEDLER as the six-tuple

$$P = (I, G, U, S, F, *),$$

with events $U \subset \mathbb{R}^3$, states S , a feature map $F : S \rightarrow \mathbb{R}^3$, and the resolution kernel

$$G(v, r) = e^{-\gamma \|F(v) - r\|}.$$

The inclination of a state is its historical prior plus the weighted match, $I(v) = \eta(v) + w G(v, r)$. The operator $*$ grows and prunes the state set: a new state is spawned when the relaxed network's order parameter falls below a novelty threshold, $R < \theta$. The Act is the transition $A(I, S_t) \rightarrow S_{t+1}$. In this guide's reading, Natural Justice is an admissibility predicate on the passage from intent to Act: a transition fires only if $\text{NJ}(\text{intent}, S_t)$ holds.

Kernels and closures, as recorded

The record gives the CEM kernel and the AyeAI Triad's closure as tuples, and they are reproduced here as given; the expansions of their letters are not part of the retrieved record:

$$\mathcal{E} = (E, \mathcal{C}, \Pi, W), \quad \text{AyeAM} = \langle S, R, C \rangle, \quad \text{AyeAI} = \langle M, I, G \rangle, \quad \text{AyeCNSe} = \langle T, Ch, \Sigma \rangle.$$

Gates and evidence

A gate is a function $g : \mathcal{A} \rightarrow \{\text{pass}, \text{fail}, \text{unjudged}\}$ on artifacts, computed by running it. An issue is done exactly when

$$g(a) = \text{pass} \wedge \text{merged}(a) \wedge (\text{published}(a) \Rightarrow \text{receipt}(a)).$$

VGC's rule is that a worker's report $r(a)$ is not an argument of g : the gate is computed from the artifact, never read from the claim.

Provenance

A repository's manifest is the list $m = [(p_i, H(f_i))]_i$ of every tracked path with the SHA-256 of its file. Sealing commits the digest $d = H(m)$ to OpenTimestamps, whose calendars aggregate many digests in a Merkle tree anchored in a Bitcoin block; the proof for one digest among N is a path of length $\lceil \log_2 N \rceil$, and verifying it recomputes the root. The four systems are ordered by implication:

$$\text{mint}(a) \Rightarrow \text{attest}(a) \Rightarrow \text{clear}(a) \Rightarrow \text{seal}(a).$$

The seed space and the reading paths

FAKIR seeds a task from a domain node, an AGI layer and a language: $S = D \times L \times \Lambda$, with $|D| = 1,505$ nodes across ISIC, ISCO and ISCED, $|L| = 10$ layers and $|\Lambda| = 7,867$ languages, so

$$|S| = 1,505 \times 10 \times 7,867 = 118,398,350.$$

A cross-domain task is seeded from a set of nodes, a subset of D , rather than one.

A reader view is a sequence of chapters $(c_1, \dots, c_k)$, a walk on the chapters. A chapter of w words is estimated at $t(c) = \max(1, \text{round}(w/200))$ minutes, and a path at $\sum_j t(c_j)$.

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